LEARNING NAVIGATOR

Mobile Educational Application

Week 1 — Application Proposal & Requirements Document

Internship Name: Mobile App Development Using Flutter

Project Name: Learning Navigator App

Team Number: 11

Team Leader: Ansa Naseer

Date:08.06.2026

1. Introduction

Learning Navigator is a mobile educational application built with Flutter that helps learners follow structured, goal-oriented learning programs in technology and professional skill domains. The app brings learning paths, curated resources, progress tracking, events, and achievements together in one place — removing the confusion of scattered content and giving learners a clear path forward.

2. App Purpose & Objectives

2.1 Problem Statement

Most learners struggle to find a clear starting point when learning a new skill. Educational content is spread across YouTube, blogs, courses, and documentation with no structure connecting them. There is no single place to track progress, stay motivated, or know what to learn next.

2.2 Purpose

Learning Navigator solves this by providing a structured, step-by-step learning experience inside a single mobile app. Learners follow organized programs, complete skills in sequence, access quality resources, participate in events, and earn recognition for their progress — all without leaving the platform.

2.3 Objectives

1. Provide structured learning paths that guide learners from beginner to advanced level.
2. Enable real-time progress tracking across enrolled programs and individual skills.
3. Offer curated, organized resources by topic and difficulty.
4. Motivate learners through a badge and achievement reward system.
5. Support community engagement through events, workshops, and webinars.
6. Give administrators tools to manage content, monitor activity, and analyze platform usage.

3. Target Audience

3.1 Learners

The primary users of Learning Navigator are individuals who want to learn technology or professional skills in a structured way. This includes:

- University students looking to supplement coursework with practical skill-building
- Self-learners who need structure and direction for independent study
- Beginners entering technology fields who do not know where to start
- Working professionals seeking to upskill or transition into a new domain

3.2 Administrators

Administrators manage the platform and its content. Their responsibilities include creating and updating learning paths, organizing events, reviewing learner feedback, and monitoring engagement through analytics dashboards.

4. Key Features

The table below summarizes the core features of Learning Navigator and their intended users.

Feature	Description	User
Authentication	Secure registration, login, and role-based access	Learner / Admin
Home Dashboard	Progress overview, active paths, quick actions	Learner
Learning Paths	Browse and enroll in structured learning programs	Learner
Skill Roadmap	Visual step-by-step node map per learning path	Learner
Resources	Curated materials (video, article, docs) per skill	Learner
Progress Tracking	Per-path and per-skill completion tracking	Learner
Events	Browse, register for workshops and webinars	Learner / Admin
Notifications	Event reminders and milestone alerts	Learner
Achievements	Badge and milestone reward system	Learner
User Profile	Personal info, learning history, badge display	Learner
Admin Dashboard	Analytics, user metrics, content management	Admin

5. Functional Requirements

Functional requirements describe what the application must do. These are organized by module.

5.1 Authentication

- Users can register with name, email, and password.
- Registered users can log in using email and password.
- The system supports role-based access — Learner and Administrator.
- A password reset option is available via email.
- The app maintains the user session across launches.

5.2 Dashboard

- The dashboard displays a personalized greeting and active learning path summary.
- Progress metrics are shown including completed skills and enrolled paths.
- Quick-action buttons provide one-tap access to key sections.
- Upcoming events and recent notifications appear as summary cards.

5.3 Learning Paths

- Users can browse, search, and filter available learning paths.
- Each path shows a title, description, difficulty level, and duration.
- Users can enroll in a path or continue from their last position.
- Progress indicators are shown per enrolled path.

5.4 Roadmap & Resources

- Each learning path has a visual step-by-step skill roadmap.
- Roadmap nodes display as locked, in-progress, or completed states.
- Each skill node links to associated curated resources.
- Resources are tagged by type: video, article, exercise, or documentation.
- Learners can mark skills and resources as completed.

5.5 Events

- Users can browse upcoming workshops, webinars, and learning events.
- Learners can register for events directly within the app.
- Administrators can create, edit, and cancel events.

5.6 Achievements & Notifications

- Badges are awarded when learners complete milestones or full learning paths.
- The profile screen displays all earned badges and achievement history.
- Push notifications are sent for event reminders and learning milestones.

- An in-app notification center maintains a history of all alerts.

5.7 Admin

- Administrators access a dedicated dashboard with platform-wide statistics.
- Admins can create, edit, and delete learning paths and roadmap content.
- Admins can view learner engagement reports and review feedback.

6. Non-Functional Requirements

Performance

- The app shall launch and display the home screen within 3 seconds on a standard Android device.
- All screen transitions shall complete within 500 milliseconds.

Usability

- A first-time user shall be able to navigate the app without any training or documentation.
- The interface shall follow Material Design 3 guidelines for consistency.
- All interactive elements shall meet a minimum touch target size of 48x48 dp.

Security

- All data transmission shall be encrypted using HTTPS/TLS.
- Passwords shall be stored securely using Firebase Authentication.
- Role-based access control shall prevent learners from accessing admin functions.

Reliability

- The app shall handle network interruptions gracefully with clear offline messages.
- User progress data shall sync to the server and not be lost on app closure or crash.

Compatibility

- The app shall support Android API level 21 (Android 5.0) and above.
- The UI shall be responsive across common phone screen sizes.

Maintainability

- The codebase shall follow clean architecture principles to support future development.
- All source code shall be version-controlled on GitHub with clear commit messages.

7. User Journey & Workflow

7.1 New Learner

A new user downloads the app, registers an account, and lands on the Home Dashboard. They browse available learning paths, select one that matches their goals, and enroll. They follow the skill roadmap in order — studying resources, completing skills, and advancing through the path. As they hit milestones, they earn achievement badges visible on their profile.

Download App → Register → Dashboard → Browse Paths → Enroll → Follow Roadmap → Complete Skills → Earn Badges

7.2 Returning Learner

A returning learner opens the app and is taken straight to their Dashboard. They review their progress, tap Continue on an active path, pick up from their last roadmap node, complete the next skill, and check upcoming events for a relevant workshop to register for.

Open App → Auto Login → Dashboard → Continue Path → Next Skill → Mark Complete → Browse Events → Register

7.3 Administrator

An administrator logs in and accesses the Admin Dashboard, which shows platform-wide metrics. From there, they manage learning path content, review learner activity, publish new events, and check submitted feedback to improve the platform over time.

Admin Login → Dashboard → View Analytics → Manage Content → Create Events → Review Feedback

8. Navigation Flow

The app uses a bottom navigation bar (Home, Paths, Events, Profile) that persists across all main screens. The overall navigation hierarchy is as follows:

```
Splash Screen
   →  Login / Registration
        →  Home Dashboard  (Bottom Nav: Home | Paths | Events | Profile)
             →  Learning Paths List
                  →  Learning Path Details  (Tabs: Roadmap | Skills | Resources)
                       →  Individual Resource View
             →  Events List  →  Event Details  →  Registration
             →  Notifications Center
             →  User Profile  →  Achievements  /  Settings
   →  Admin Login  →  Admin Dashboard  →  Content Management  /  Analytics
```

9. Wireframe Designs

The following wireframes were designed in Figma and represent the initial screen layouts for Learning Navigator. Insert your Figma screenshots into the placeholders below.

PROTOTYPE LINK:

The initial wireframe structure was generated using AI-assisted design tools to accelerate the ideation process. All screens, navigation flow, features, and content were subsequently reviewed, customized, and refined by the project team to meet the specific requirements of the Learning Navigator application.

Wireframe Link: https://www.figma.com/make/Y3nWiwIfkEbosLBvpJr5Tf/Convert-Resume-Analyzer-to-Learning-App?t=WheeFgeROkG512BA-1&preview-route=%2Fhome

Learner Login Page:

Enables learners to securely log in and access their personalized learning dashboard, courses, and progress tracking.

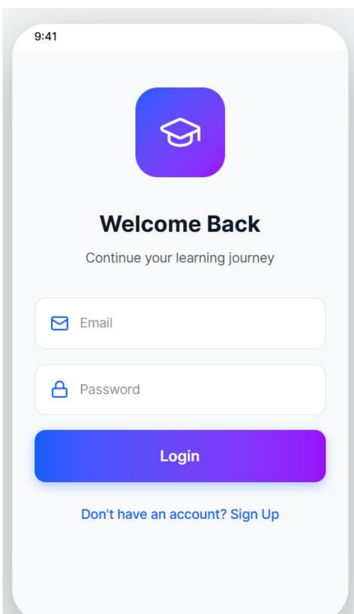


Fig: Login Page

Learner Home Page

Enables learners to explore learning paths, access recommended resources, and track their learning progress from a centralized dashboard.

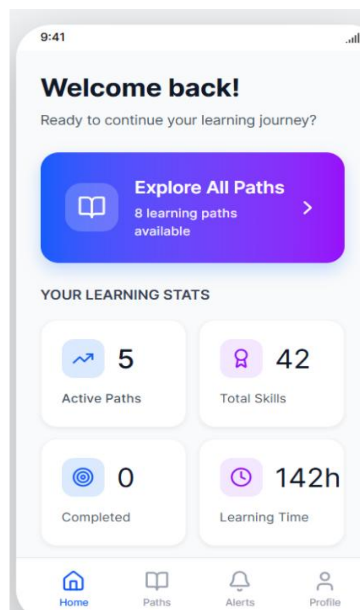


Fig: Home Page

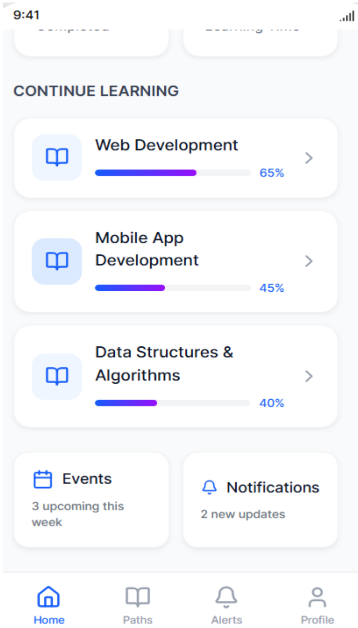


Fig: Home Page

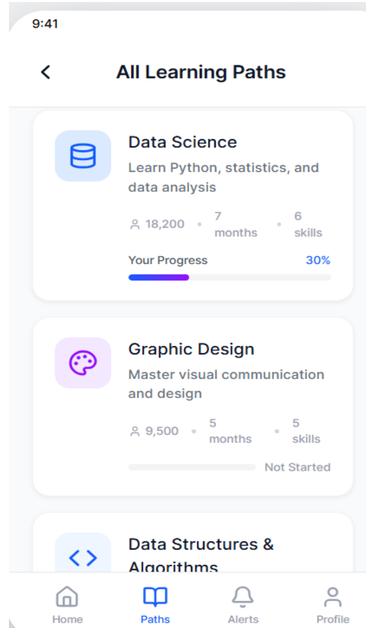


Fig: Path Page

Learner Path Page:

Enables learners to view and follow a structured learning roadmap, guiding them through each stage of their chosen skill or career path.

Learner Alert Page:

Displays a personalized learning roadmap with step-by-step milestones, helping learners progress toward their chosen skill or career goal.

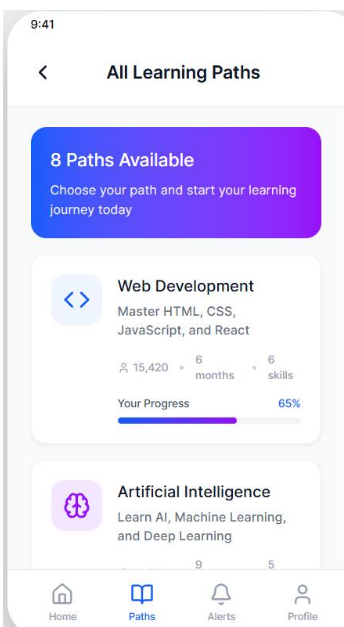


Fig: Path Page

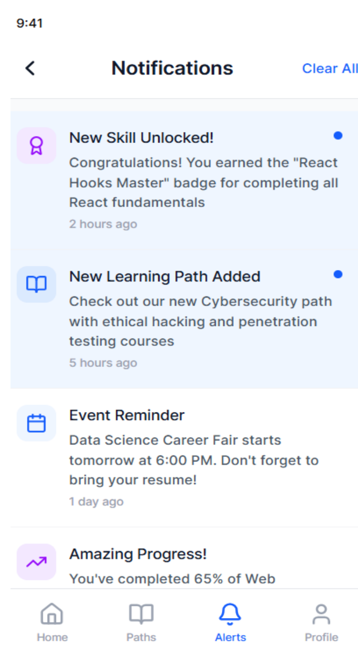


Fig: Alert Page

Learner Profile Page:

Enables learners to view and manage their personal information, learning goals, achievements, and account settings.

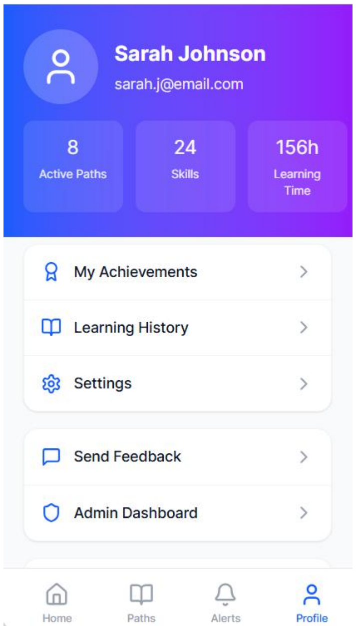


Fig: Profile Page

Admin Login Page:

Enables authorized administrators to securely access the admin dashboard and manage learners, content, and platform activities.

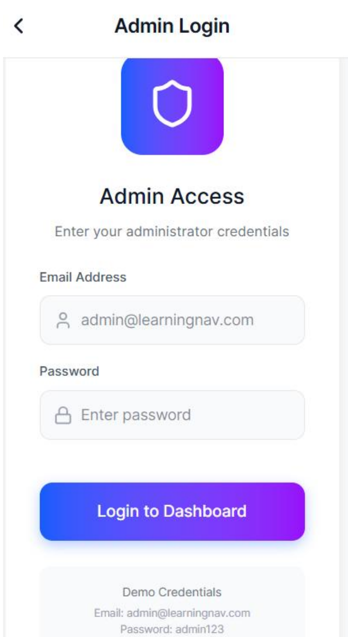


Fig: Admin Access

Admin Dashboard:

Enables administrators to monitor platform activity, manage learners and content, and access key analytics from a centralized control panel.

Fig: Admin Dashboard

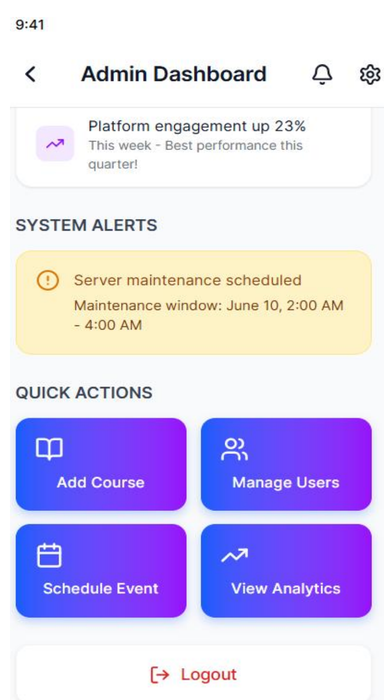


Fig: Admin Dashboard

Admin Notification Page:

Enables administrators to view, manage, and send important announcements, updates, and alerts to learners.

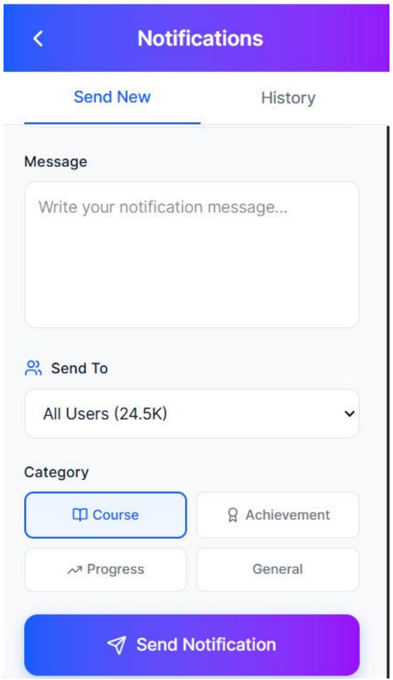


Fig: Admin Notification

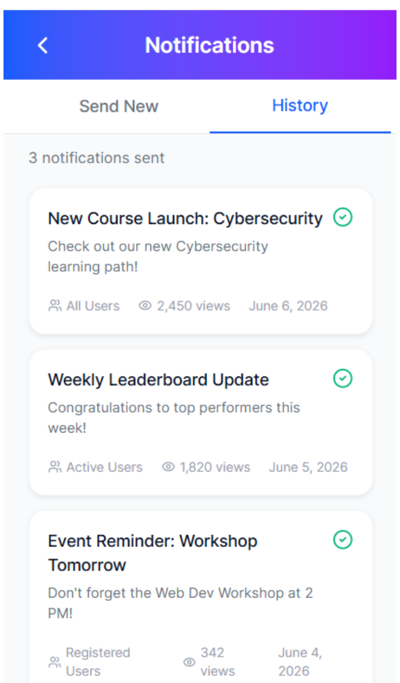


Fig: Admin Notification

Setting Page:

Enables users to manage account preferences, update personal information, adjust app settings, and control security options.

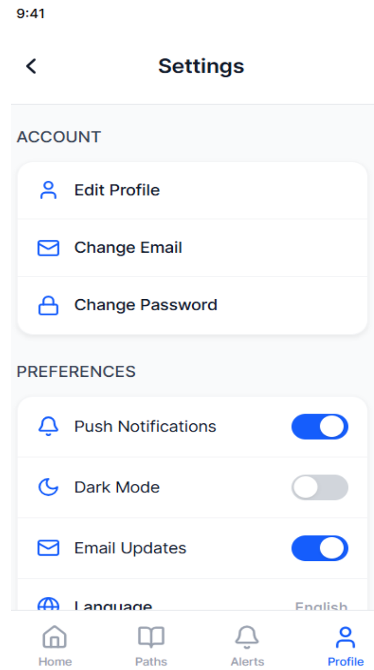


Fig: Settings

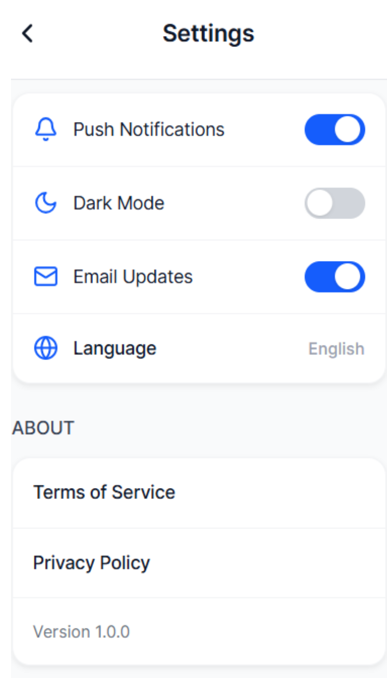
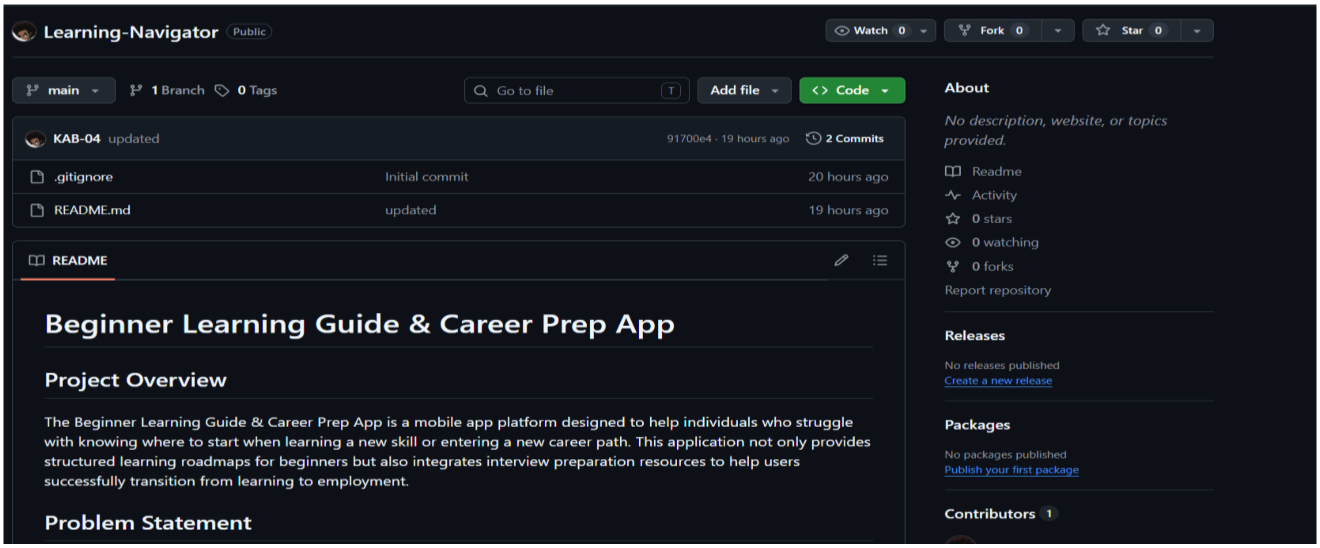
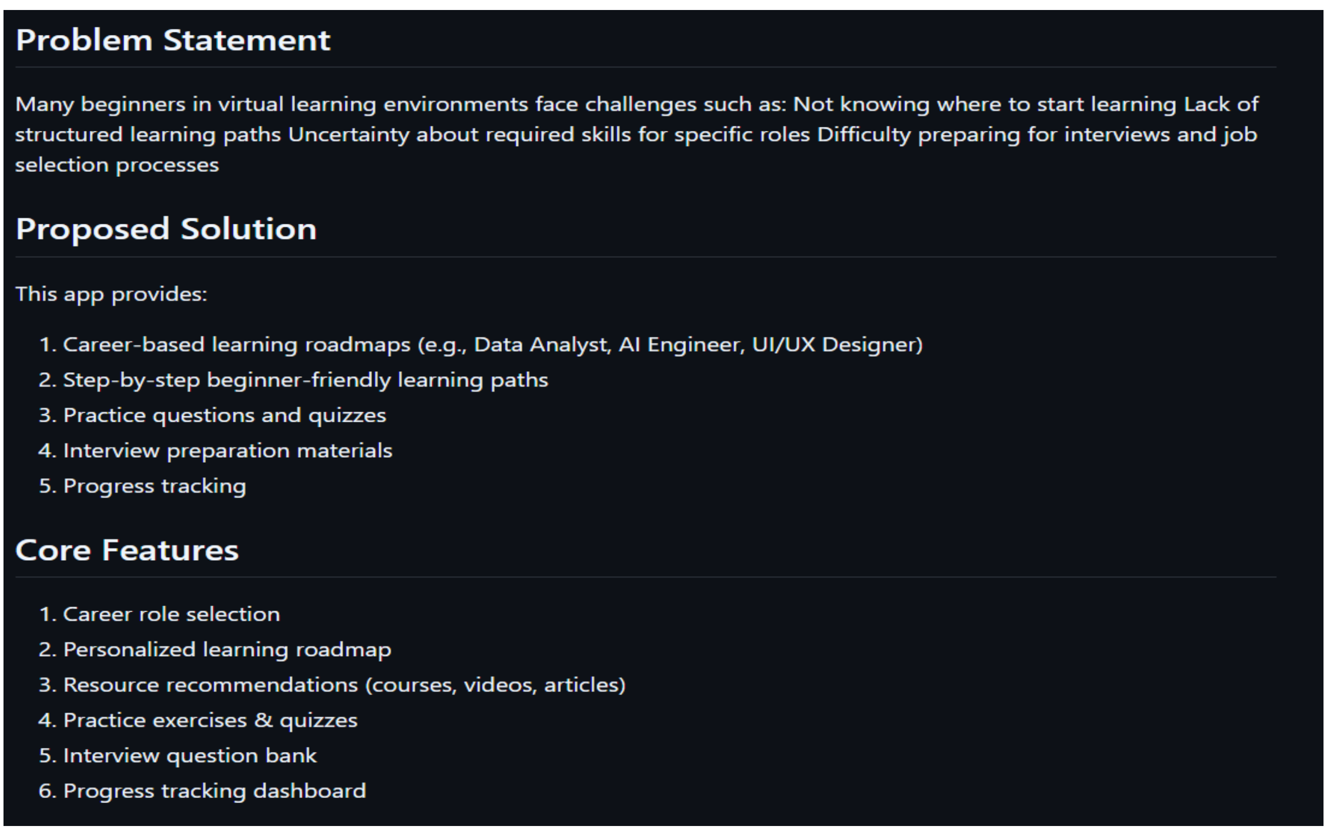


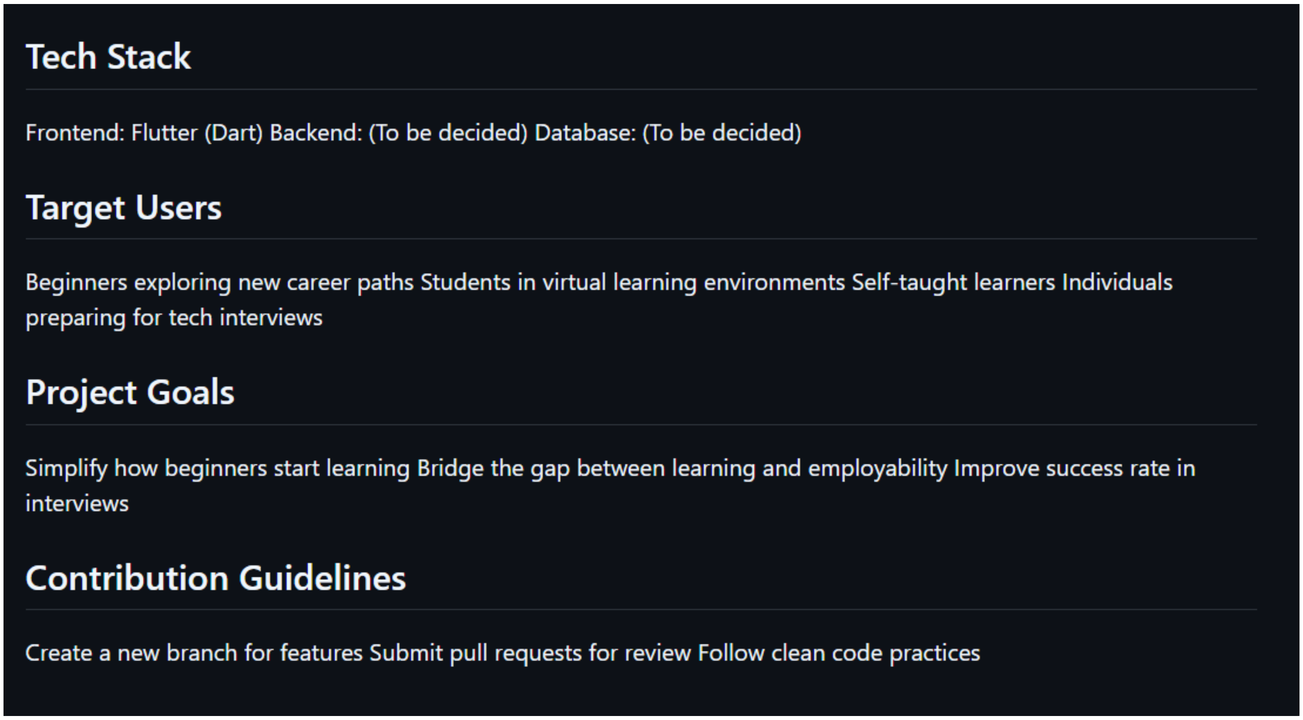
Fig: Settings

10.GitHub Setup

The Learning Navigator app uses GitHub for version control and collaboration. The repository contains all source code, documentation, and project files. The project follows a branch-based workflow where the main branch holds the stable version and feature branches are used for development. Changes are reviewed through pull requests before merging. GitHub is also used for tracking issues and managing tasks, ensuring smooth collaboration and organized development.







Tech Stack

Frontend: Flutter (Dart) Backend: (To be decided) Database: (To be decided)

Target Users

Beginners exploring new career paths Students in virtual learning environments Self-taught learners Individuals preparing for tech interviews

Project Goals

Simplify how beginners start learning Bridge the gap between learning and employability Improve success rate in interviews

Contribution Guidelines

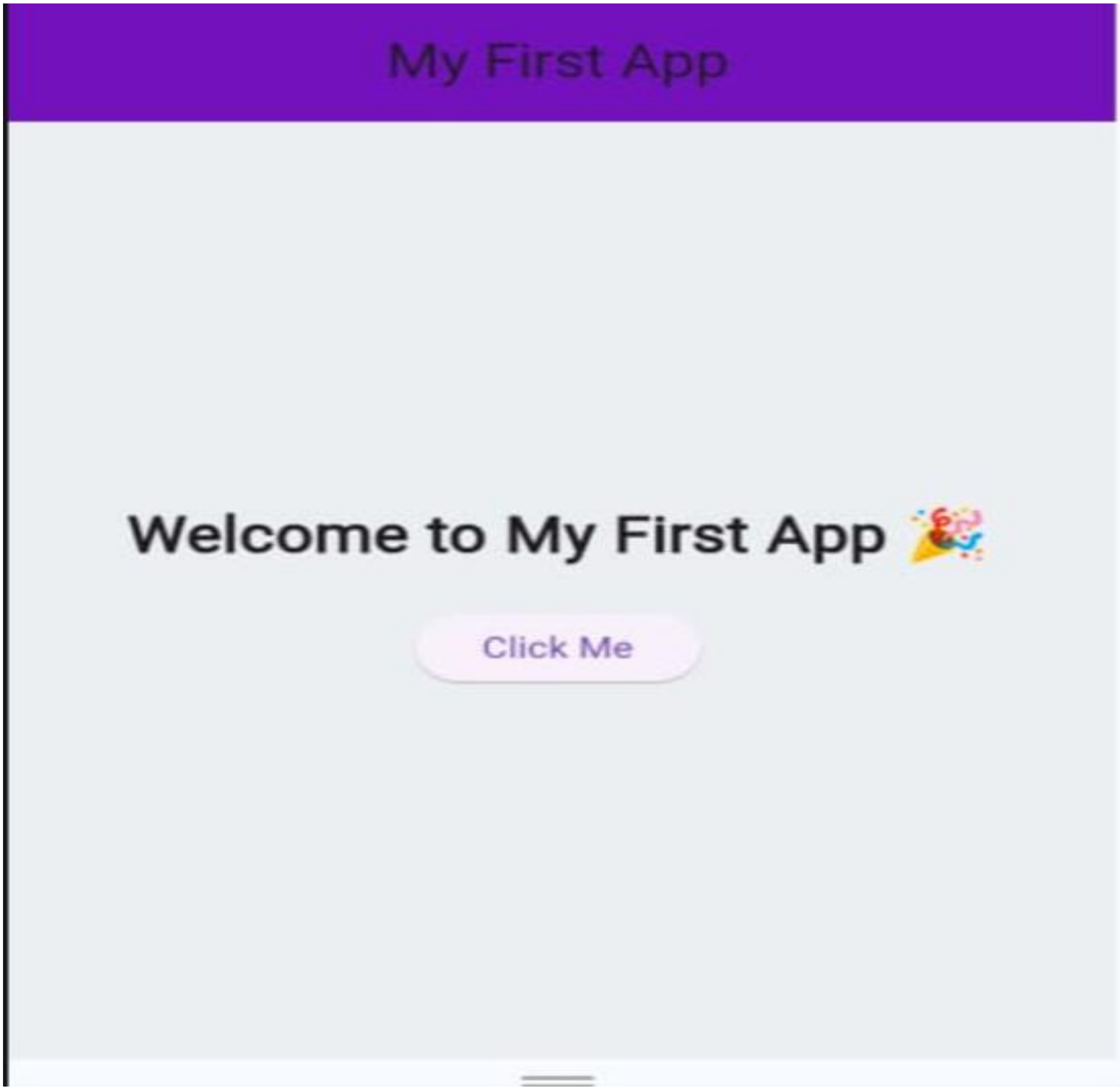
Create a new branch for features Submit pull requests for review Follow clean code practices

11.Flutter Setup and Demo App

Flutter was installed and configured with the required SDK, Android Studio, and emulator. The environment was verified using Flutter tools to ensure proper setup. A demo Flutter application was then created and executed successfully to validate the installation. The demo app confirmed that the development environment was ready for building and testing the Learning Navigator application.

```
The default interactive shell is now zsh.
To update your account to use zsh, please run `chsh -s /bin/zsh`.
For more details, please visit https://support.apple.com/kb/HT208050.
Waseems-MacBook-Air:~ waseemcomputernwl$ echo 'export PATH="$PATH:/Users/waseemcomputernwl/Downloads/flutter/bin"' >> ~/.bash_pr
Waseems-MacBook-Air:~ waseemcomputernwl$ source ~/.bash_profile
Waseems-MacBook-Air:~ waseemcomputernwl$ flutter doctor

  ┌─────────────────────────────────────────────────────────┐
  │ A new version of Flutter is available!                  │
  │                                                         │
  │ To update to the latest version, run "flutter upgrade". │
  └─────────────────────────────────────────────────────────┘
Doctor summary (to see all details, run flutter doctor -v):
[✓] Flutter (Channel stable, 3.38.8, on macOS 12.7.6 21H1320 darwin-x64, locale
    en-PK)
[!] Android toolchain - develop for Android devices (Android SDK version 36.1.0)
    ! Some Android licenses not accepted. To resolve this, run: flutter doctor
      --android-licenses
[✗] Xcode - develop for iOS and macOS
    ✗ Xcode installation is incomplete; a full installation is necessary for iOS
      and macOS development.
      Download at: https://developer.apple.com/xcode/
      Or install Xcode via the App Store.
      Once installed, run:
        sudo xcode-select --switch /Applications/Xcode.app/Contents/Developer
        sudo xcodebuild -runFirstLaunch
    ✗ CocoaPods not installed.
        CocoaPods is a package manager for iOS or macOS platform code.
        Without CocoaPods, plugins will not work on iOS or macOS.
        For more info, see https://flutter.dev/to/platform-plugins
      For installation instructions, see
      https://guides.cocoapods.org/using/getting-started.html#installation
[✓] Chrome - develop for the web
[✓] Connected device (2 available)
[✓] Network resources

! Doctor found issues in 2 categories.
The Flutter CLI developer tool uses Google Analytics to report usage and
diagnostic
data along with package dependencies, and crash reporting to send basic crash
reports. This data is used to help improve the Dart platform, Flutter framework,
and related tools.

Telemetry is not sent on the very first run. To disable reporting of telemetry,
run this terminal command:

    flutter --disable-analytics

If you opt out of telemetry, an opt-out event will be sent, and then no further
information will be sent. This data is collected in accordance with the Google
Privacy Policy (https://policies.google.com/privacy).

Waseems-MacBook-Air:~ waseemcomputernwl$ flutter doctor --android-licenses
/Users/waseemcomputernwl/Library/Android/sdk/cmdline-tools/latest/bin/sdkmanager: line 173: test: : integer expression expected
[=======================================] 100% Computing updates...
6 of 7 SDK package licenses not accepted.
Review licenses that have not been accepted (y/N)? y

1/6: License android-googletv-license:
---------------------------------------
Terms and Conditions
```

```
\ccept? (y/N): y
\ll SDK package licenses accepted

Vaseems-MacBook-Air:~ waseemcomputernwl$ cd ~/Desktop
Vaseems-MacBook-Air:Desktop waseemcomputernwl$ flutter create first_app
Creating project first_app...
Resolving dependencies in `first_app`... (1.3s)
Downloading packages... (10.9s)
Got dependencies in `first_app`.
Wrote 130 files.

\ll done!
You can find general documentation for Flutter at: https://docs.flutter.dev/
Detailed API documentation is available at: https://api.flutter.dev/
If you prefer video documentation, consider:
https://www.youtube.com/c/flutterdev

In order to run your application, type:

  $ cd first_app
  $ flutter run

Your application code is in first_app/lib/main.dart.

Vaseems-MacBook-Air:Desktop waseemcomputernwl$ cd first_app
Vaseems-MacBook-Air:first_app waseemcomputernwl$ flutter devices
Found 2 connected devices:
  macOS (desktop) • macos   • darwin-x64      • macOS 12.7.6 21H1320 darwin-x64
  Chrome (web)    • chrome • web-javascript • Google Chrome 148.0.7778.216

No wireless devices were found.

Run "flutter emulators" to list and start any available device emulators.

If you expected another device to be detected, please run "flutter doctor" to
diagnose potential issues. You may also try increasing the time to wait for
connected devices with the "--device-timeout" flag. Visit
https://flutter.dev/setup/ for troubleshooting tips.
Vaseems-MacBook-Air:first_app waseemcomputernwl$ flutter run -d chrome
Launching lib/main.dart on Chrome in debug mode...
Waiting for connection from debug service on Chrome...             29.9s

Flutter run key commands.
r Hot reload. 🔥🔥🔥
R Hot restart.
h List all available interactive commands.
d Detach (terminate "flutter run" but leave application running).
c Clear the screen
q Quit (terminate the application on the device).

This app is linked to the debug service: ws://127.0.0.1:49361/blruoCzaA-s=/ws
Debug service listening on ws://127.0.0.1:49361/blruoCzaA-s=/ws
A Dart VM Service on Chrome is available at: http://127.0.0.1:49361/blruoCzaA-s=
The Flutter DevTools debugger and profiler on Chrome is available at:
http://127.0.0.1:49361/blruoCzaA-s=/devtools/?uri=ws://127.0.0.1:49361/blruoCzaA
-s=/ws
Starting application from main method in: org-dartlang-app:/web_entrypoint.dart.
Application finished.
Vaseems-MacBook-Air:first_app waseemcomputernwl$
```


My First App
Welcome to My First App 🎉
Click Me